

A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Specialist techniques & statistical analysis - Set 1

Mark scheme

AQA spec 3.7.2.3; Research methods; textbook pp. 402-407, 420-423

Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>2 marks for each point (partial credit for a partly correct point).</i> • Image databases: identifying individuals (tiger stripes, whale shark spots, dolphin fins) to find territory size, movements, lifespan, social groups - few species have unique recognisable features (2) • eDNA: detecting the presence of a species from DNA shed into water (great crested newt, invasive carp) - shows presence not abundance; contamination / DNA transport by water (2) • Geolocator tags: tracking small birds on migration where transmitters are too heavy - only record light and time so position is approximate; animal must be recaptured to download data (2)	6
01.2	<i>1 mark for each point.</i> • Microwaves emitted by the satellite reflect off the surface giving data on wave height, wind velocity, sea surface altitude and oil slicks (which smooth the surface) (1) • Microwaves pass through cloud so monitoring is not interrupted by weather (1)	2
01.3	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Droppings: diet, gender, territory size (1) • Owl pellets: diet (1) • Feeding marks (chewed hazel nuts - dormouse), tracks / footprints, nests / burrows, scratching posts (territory) (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 402-407; AQA spec 3.7.2.3.	

Question	Marking guidance	Marks
02.1	<i>1 mark for each point.</i> - The scatter / variability of results around each mean; about 68% of values lie within one SD of the mean (1) - The SDs are large relative to the difference in means (1.9 g) so the ranges overlap greatly (1)	2
02.2	<i>1 mark for each point.</i> - t-test (1) - Comparing two means of measured (continuous) data (1)	2
02.3	<i>1 mark for each point.</i> - There is a 30% probability that the difference occurred by random chance (1) - Above the 0.05 (5%) level so the difference is not significant; the null hypothesis (no difference) is accepted (1)	2
02.4	<i>1 mark for each point.</i> - Two correlated variables may both be caused by a third variable that was not investigated (1) - Significance only shows the result is unlikely to be chance; a mechanism / controlled experiment is needed for causation (1)	2
02.5	<i>1 mark for each point.</i> - Increase the number of replicates (reduces the effect of variability) (1) - Standardise other variables (soil, water, light) more closely to reduce scatter (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 420-423; AQA spec 3.7.2.3.	
03.1	<i>1 mark for each point.</i> - Spearman's rank correlation coefficient (1) - Chi-squared test (frequencies / counts in categories) (1) - t-test (two means of measured data) (1) - Spearman's rank correlation coefficient (1)	4
03.2	<i>1 mark.</i> Chi-squared compares frequencies / counts in categories; it cannot be used for continuously variable measured data (1)	1
03.3	<i>1 mark for each point.</i> - The value (ignoring sign) exceeds the critical value so the correlation is significant at the 5% level (1) - The negative sign shows a negative correlation: one variable decreases as the other increases (1) - There is less than a 5% probability the correlation arose by chance; reject the null hypothesis (but correlation is not proof of causation) (1)	3
03.4	<i>1 mark for each point.</i> - Show the standard deviation / range of the original values around each mean (1) - If the bars of two means overlap greatly the difference is probably not significant; little overlap suggests a significant difference (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 422-423; AQA spec 3.7.2.3.	

Question	Marking guidance	Marks
04.1	<i>1 mark for each point.</i> - Each regional population has its own unique gene pool / DNA profile (1) - DNA from seized timber or ivory is compared with the database of regional populations (1) - This identifies where the material came from, showing whether it was legally harvested and where poaching occurs (1)	3
04.2	<i>1 mark for each point.</i> - Radio / GPS transmitters use radio waves - work in air and fresh water, giving positions (GPS stores or transmits location) (1) - Radio waves do not travel through seawater so acoustic transmitters (sound) are used for marine animals, giving real-time positions (1)	2
04.3	<i>Up to 3 marks in total; award marks in proportion to the completeness of each point.</i> - Drones / ROVs, AUVs (eg Autosub), balloons, aircraft, satellites, animals (tags) (up to 2) - Drones: cheap, low disturbance, high-resolution images of areas hard to reach (cliffs, canopy, ice), repeated surveys; infrared cameras detect animals at night (up to 2)	3
04.4	<i>1 mark for each point.</i> - Cameras record automatically (visible light by day, infrared at night) without researchers being present (1) - So animals behave naturally and nests are not deserted; monitoring continues 24 hours (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 404-406; AQA spec 3.7.2.3.	
05.1	<i>A correct final answer (with unit where asked) scores full marks. If the answer is wrong, award 1 mark for each correct step of working shown.</i> $\Sigma n(n-1) = (40 \times 39) + (30 \times 29) + (20 \times 19) + (10 \times 9) = 1560 + 870 + 380 + 90 = 2900$ (1) $N(N-1) = 100 \times 99 = 9900$ (1) $D = 9900 / 2900 = 3.41$ (accept 3.4) (1)	3
05.2	<i>1 mark for each point.</i> - Woodland A has a higher biodiversity (greater variety and more even abundance) than woodland B (1) - The index does not identify which species are present / a rare or important species may be missed; depends on identification skill and sampling method (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 402; AQA spec 3.7.2.3.	
06.1	<i>1 mark for each point.</i> - t-test (1) - The data are measured (continuous) values and two means are being compared (1) - The data are (assumed to be) normally distributed (1)	3
06.2	<i>1 mark.</i> Use the Mann-Whitney U test instead (compares medians) (1)	1

Question	Marking guidance	Marks
06.3	<i>1 mark for each point.</i> • There is a 2% probability that the difference occurred by random chance (1) • As this is less than 0.05 the difference is statistically significant / the null hypothesis is rejected (1)	2
06.4	<i>1 mark for each point.</i> • The ranges of values (mean plus or minus SD) do not overlap: woodland up to 1.6 and clearing down to 2.4 (1) • About 68% of values lie within one SD of each mean so the two data sets are clearly separated (1)	2
06.5	<i>1 mark for each point.</i> • Significance shows the difference is unlikely to be due to chance, not that a causal relationship exists (1) • Another variable (eg topography / aspect of the sites) could cause both; a controlled experiment or further evidence is needed (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 420-423; AQA spec 3.7.2.3.	
07.1	<i>1 mark for each point.</i> • Spearman's rank correlation coefficient (1) • Mann-Whitney U test (1) • Chi-squared test (1) • t-test (1)	4
07.2	<i>1 mark for each point.</i> • Chi-squared compares frequencies / counts in categories (1) • Dissolved oxygen is continuously variable measured data, so a t-test or Mann-Whitney U test is needed (1)	2
07.3	<i>1 mark for each point.</i> • There is always some remaining uncertainty / a probability that the result was produced by chance (1) • Statistics only give a level of confidence (eg 95%, 99%) that the result is significant (1)	2
07.4	<i>1 mark for each point.</i> • Bars drawn above and below each mean showing plus and minus one standard deviation (1) • If the bars of two means overlap a lot the difference is probably not significant; if they do not overlap it probably is (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 422-423; AQA spec 3.7.2.3.	